Regularized Microscopic X-ray CT Reconstruction via Flow Matching

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Abstract
Recent works in generative prior regularized image reconstruction show promising results in reducing noise and structural artifacts. In this work, we explore the feasibility of using generative priors, such as flow matching models, to regularize volumetric cone-beam computed tomography (CBCT) reconstruction for X-ray microscopy. We show such methods noticeably reduce noise and cone beam artifacts from short-scan and rotational laminography, therefore have great potential to be used in high throughput high resolution CBCT applications.

Keywords: generative model, inverse problem, regularization by denoising.

1 Introduction

Cone beam computed tomography (CBCT) is a tomographic X-ray imaging method that creates a digital volumetric representation of the internal structure of the sample with X-ray contrast, such as absorption, phase, and scattering. To achieve a high quality 3D reconstruction from 2D projections, in general, Tuy's condition needs to be satisfied [1]. In reality, most CBCT scanning trajectory does not satisfy such a condition. In addition, to reduce dose or increase throughput, novel trajectories such as short scan, sparse-view scan, and rotational laminography trajectory can result in more ill-posed inverse problems, causing more observable streaking and shading artifacts such as limited view and cone beam artifacts. Moreover, due to the quantum nature of light and imperfect detector technology, photon noise and thermal and readout noise are typically present in the acquired image, resulting in noisy reconstruction.
In this work, we experiment with a re-noising flow-based workflow to regularize the CBCT reconstruction. We focus on two types of scan trajectories, short-scan and rotational laminography, that are typically used in high-throughput or low-dose applications, which have well-recognized reconstruction artifacts.

2 Short-scan

Short-scan, also known as 180+fan scan, is a widely used circular scanning trajectory for asymmetric samples. As the name suggests, projection angles only cover 180 degree plus the fan angle range. In industry CT settings, this could be useful for anisotropic samples that need to be placed close to the source for higher geometric magnification, for example. However, for CBCT, especially in the high cone angle regimes, a short scan can result in streak artifacts. Figure 2 shows simulation results from a short-scan of a foam sample. From the filtered backprojection (FBP) reconstruction, we see expected streaking artifacts at boundary, and elongated cone beam artifacts, as well as random noise. The iterative reconstruction without regularization corrects some of the cone beam artifacts, and make the reconstruction more uniform. The noise2noise-based DeepReconIC method [2, 3] removes both noise and most sampling artifacts, but was not able to recover smeared and distorted features. Both diffusion posterior sampling-based (DPS) reconstruction [4] and the proposed flow-based reconstruction reduce noise and artifacts while recovering the structure. Empirically, the proposed flow-based method outperforms the others.

3 Rotational laminography

Another popular geometry typically used for large, flat samples is rotational laminography [5]. To scan a plate-like sample, such as a wafer or a printed circuit board, it is often advantageous to use a laminography trajectory for a few reasons. First, the sample

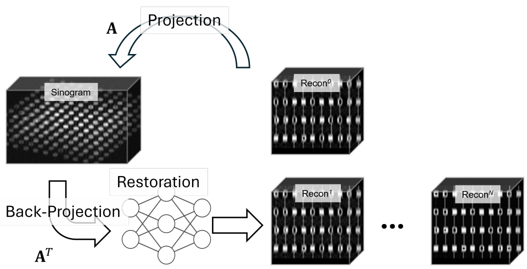

Figure 1: Proposed reconstruction workflow, that use a generative prior to regularize the reconstruction

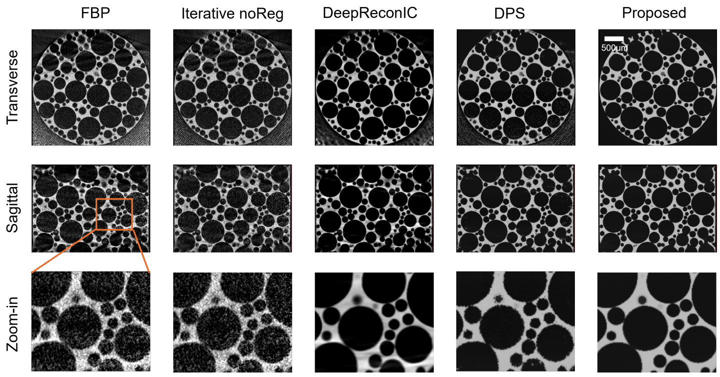

Figure 2: Reconstruction of a foam sample from a simulated short-scan

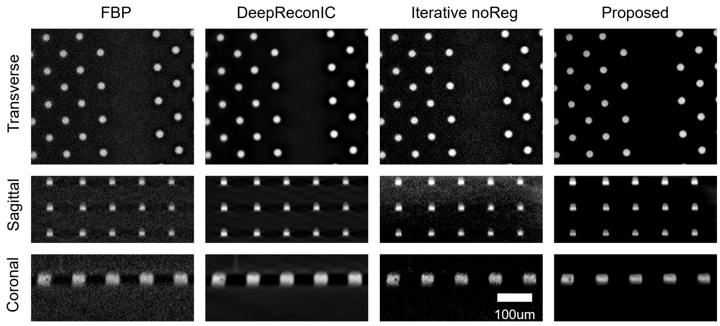

Figure 3: Reconstruction of a DRAM sample scanned with an in-house laminography prototype system

can rotate quickly and stably when laid flat. Second, the beam path penetrating the sample is relatively short. Third, the source can be placed very close to the sample for high geometric magnification. However, reconstruction from such a trajectory suffers from smearing cone-beam artifacts, as well as vignetting or a truncated field of view (FOV). Figure. 3 illustrates a DRAM sample reconstructed from measurements taken with an in-house prototyping laminography scanner [6]. We can see the aforementioned tail artifacts in the FBP reconstruction. DeepReconIC [2, 3] is able to remove noise but is unable to mitigate cone-beam artifacts. The proposed flow-based method is capable of reducing both noise and laminography artifacts.

4 Conclusion

In this work, we demonstrated via numerical simulation and experimental validation that flow matching models effectively regularize CBCT reconstructions, significantly reducing noise and artifacts in challenging trajectories such as short-scan and rotational laminography. Our results indicate that the proposed flow-based reconstruction workflow excels in artifact removal and feature recovery. Future work will focus on investigating efficient, robust frameworks, such as ADMM-inspired workflows [7].

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